Parul® University
Vadodara, Gujarat

NAAC GRADE A++

Information and Communication Technology

UNIT-6
VISUALIZATION

Study Guide

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Data visualization

Data visualization is the process of representing data in a graphical or visual format so that patterns, trends and insights can be easily understood.

Data visualization techniques such as box plots, histograms, scatter plots, and heat maps help understand data patterns. It is essential for identifying patterns, trends and outliers, enabling faster, data-driven decisions while making complex insights accessible to non-technical stakeholders.

Tableau enables interactive visualization, outlier detection identifies abnormal values and data balancing improves machine learning performance by handling imbalanced datasets.

Importance of Data Visualization

1. Simplifies complex data
2. Helps in better decision making
3. Detects patterns and trends
4. Identifies outliers and anomalies
5. Improves understanding of large datasets

Types of Data Visualization Techniques:

1. **Scatter Plots:** Ideal for analyzing relationships between two numerical variables.
2. **Histograms:** Used for understanding the distribution of numerical data.
3. **Heatmaps:** Effective for visualizing relationships between multiple variables using color.
4. **Box Plots:** Used to show the distribution of data points and identify outliers.
5. **Dashboards:** Provide real-time, interactive insights for monitoring key performance indicators (KPIs)

Importance of Data Visualization

Data visualization is essential for understanding and communicating information effectively. Here are some key reasons why it's important:

1. **Simplifies Complex Data:** It turns large and complicated data into visual formats like charts and graphs, making the information easier to understand.
2. **Reveals Patterns and Trends**: It helps identify trends, relationships and patterns that are not easily seen in raw data or tables.

3. **Saves Time:** Visuals allow quicker interpretation of data, helping users spot key information at a glance instead of manually scanning through numbers.
4. **Improves Communication:** It makes it easier to explain data insights to others, especially those who may not be familiar with the technical details.
5. **Tells a Clear Story**: Data visuals guide the audience through the information step-by-step, making it easier to reach conclusions and make informed decisions.

Real-World Use Cases

Data visualization is used across various industries to improve decision-making and drive results. Here are a few examples:

1. **Business Analytics:** Used to monitor company performance, track KPIs and make data-driven decisions by visualizing trends, sales and customer metrics.
2. **Healthcare:** Helps in analyzing patient records, tracking disease outbreaks and managing hospital operations through easy-to-read charts and dashboards.
3. **Sports:** Used to visualize player statistics, team performance and match outcomes, helping coaches and analysts improve strategies and training plans.
4. **Retail and E-commerce:** Enables tracking of sales, customer preferences and inventory levels, helping businesses adjust stock and marketing efforts effectively.

Key Aspects of Data Visualization:

- **Common Types**: Charts and graphs, such as bar charts, line charts, scatter plots, histograms, and box plots, are used for analysis, along with maps for spatial data.
- **Key Techniques**: Techniques include basic plots, multi-dimensional visuals, and interactive dashboards, which are applied across machine learning, business intelligence, and data mining.
- **Tools for Visualization**: Popular tools include Python libraries (Matplotlib, Seaborn, Plotly), R, Tableau, Power BI, and Google Data Studio.
- **Importance**: Visualizations help quickly interpret large datasets, allowing for quick identification of data trends, correlations, and outliers.

Challenges

1. **Data Quality**: Accuracy of visualizations depends on the quality of the data. If the data is inaccurate or incomplete, the insights from the visualization will be misleading.
2. **Over-Simplification**: Simplifying data too much can lead to important details being lost like using

a pie chart that oversimplifies complex relationships between categories.

3. **Choosing the Right Visualization**: Using the wrong type of visualization can distort the message. For example, a pie chart might not work well with many categories which leads to confusion.
4. **Overload of Information**: Too much information in a visualization can overwhelm viewers. It's important to focus on key data points and avoid clutter.

Box Plot (Box-and-Whisker Plot)

A box plot displays distribution of numerical data, showing median, quartiles and outliers. It's useful for understanding variability and detecting unusual values. A Box Plot (also called a Box-and-Whisker Plot) is a graphical method used in data science and statistics to visualize the distribution, spread and variability of numerical data. It provides a summary of data using five important statistical values and helps in identifying outliers, skewness and dispersion in a dataset. A box plot is especially useful when analyzing large datasets because it presents a compact summary of data distribution.

It helps compare distributions between groups with ease.

- Multiple groups effectively.
- Identify outliers quickly and visually.
- Reveal data symmetry or skewness at a glance.
- Show variability clearly even for large datasets.
- Provide compact summaries of data using quartiles and range.
- Useful for comparing multiple groups or categories.

Real Life Example:

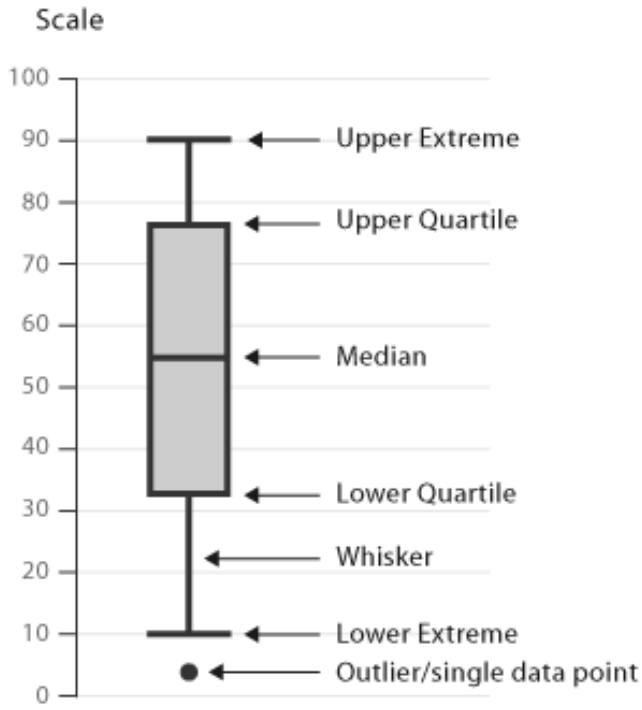
1. **Healthcare:** Detect abnormal patient reports.
2. **Finance:** Analyze stock prices and risks.
3. **Education:** Analyze student performance.
4. **Business:** Compare sales across regions.

A box plot (or box-and-whisker plot) in data science is a standardized, compact graphical tool used to visualize the distribution of numerical data based on a 5 points.

five-number summary:

1. minimum

2. first quartile (Q1)
3. median (Q2)
4. third quartile (Q3)
5. maximum



Interquartile Range (IQR)

The box in a box plot represents the Interquartile Range (IQR).

$$IQR = Q3 - Q1$$

It covers the middle 50% of the data, making it a strong measure of spread that is resistant to extreme values.

Identifying Outliers in a Box Plot

Outliers are values that lie outside the upper and lower limits:

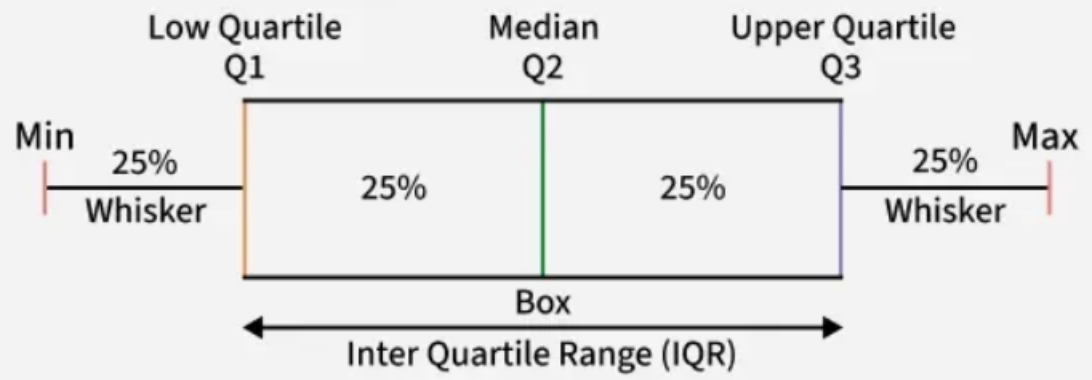
Lower Limit = **Q1 − 1.5 × IQR**

Upper Limit = **Q3 + 1.5 × IQR**

- Any value below the lower limit or above the upper limit is treated as an outlier and shown as a separate point.
- The minimum and maximum whiskers represent the closest values within these limits.

Components of a Box Plot

A Box Plot visually represents the five-number summary of a dataset:



The box plot shown in the above diagram is a perfect plot with no skewness. The plots can have skewness and the median might not be at the center of the box.

- **Minimum:** The smallest non-outlier value shown at the lower whisker end.
- **First Quartile (Q1):** The value below which 25% of the data lies.
- **Median (Q2):** The middle value that splits the data into two equal halves.
- **Third Quartile (Q3):** The value below which 75% of the data lies.
- **Maximum:** The largest non-outlier value shown at the upper whisker end.

How to Create a Box Plot

Consider the runs scored by a cricket team in 12 matches: 100, 120, 110, 150, 110, 140, 130, 170, 120, 220, 140, 110

Step 1: Arrange Data in Ascending Order

- 100, 110, 110, 110, 120, 120, 130, 140, 140, 150, 170, 220

Step 2: Find Quartiles and Median

- Median (Q2)=120+1302=125Median (*Q*2)=2120+130=125
- Q1=Median of first 6 values=110+1102=110*Q*1=Median of first 6 values=2110+110=110
- Q3=Median of last 6 values=140+1502=145*Q*3=Median of last 6 values=2140+150=145

Step 3: Compute IQR

- IQR=145−110=35IQR=145−110=35

Step 4: Calculate Limits

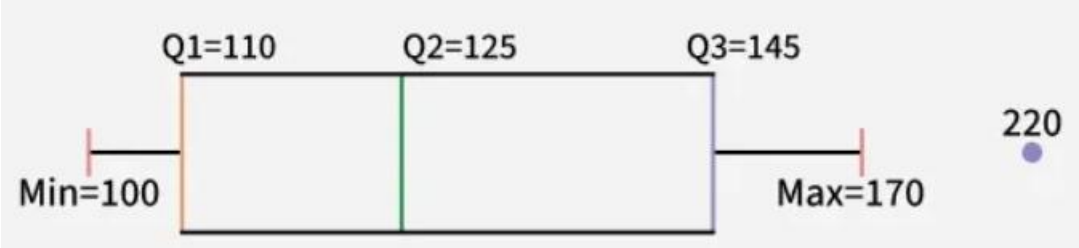
- Lower Limit=110−1.5×35=57.5Lower Limit=110−1.5×35=57.5

- Upper Limit=145+1.5×35=197.5Upper Limit=145+1.5×35=197.5

Step 5: Identify Min, Max and Outliers

- Minimum (within limit) = 100
- Maximum (within limit) = 170
- Outlier = 220

This information is then used to draw the box plot.



Advantages

- **Easy Comparison:** Allows quick comparison of distributions across multiple groups.
- **Outlier Detection:** Clearly highlights unusual or extreme values in the data.
- **Quartile-Based Summary:** Summarizes data using median and quartiles instead of raw values.
- **Resistant to Noise:** Less affected by extreme values compared to mean based visuals.
- **Compact Representation:** Presents a large amount of statistical information in a clean, compact format.

Limitation

- **No Distribution Shape:** Does not reveal whether data is normal, uniform or multi-peaked.
- **Mean Not Visible:** Shows median and quartiles but hides the average value.
- **Cannot Detect Multi-Modality:** Fails to identify multiple peaks or clusters in the data.
- **Less Useful for Small Samples:** Provides limited insights when dataset size is very small.

Difference Between Vertical and Horizontal Box Plot

Here we compare vertical and horizontal box plot

Feature	Vertical Box Plot	Horizontal Box Plot
Orientation	Box is drawn vertically along the y-axis	Box is drawn horizontally along the x-axis
Best For	Comparing numerical values across categories.	Showing distribution when category labels are long.
Readability	Can become cramped with many categories.	More readable when dataset has many labels.
Common Use	Default plot in most statistical tools.	Useful for survey data, ratings and long text labels.
Space Utilization	Requires more vertical space.	Utilizes horizontal space more efficiently.

Histogram

A Histogram is a graphical representation used in data science and statistics to display the frequency distribution of continuous numerical data.

A Histogram is a graph that represents the distribution of numerical data by dividing data into intervals called **bins (or classes)** and showing the frequency of observations in each interval.

It helps in understanding:

- Data distribution
- Frequency of occurrence
- Shape of data
- Spread and variability
- Presence of skewness and outliers

A histogram converts raw numerical data into visual form using **bars**. Unlike a bar chart, the bars in a histogram **touch each other**, indicating continuous data.

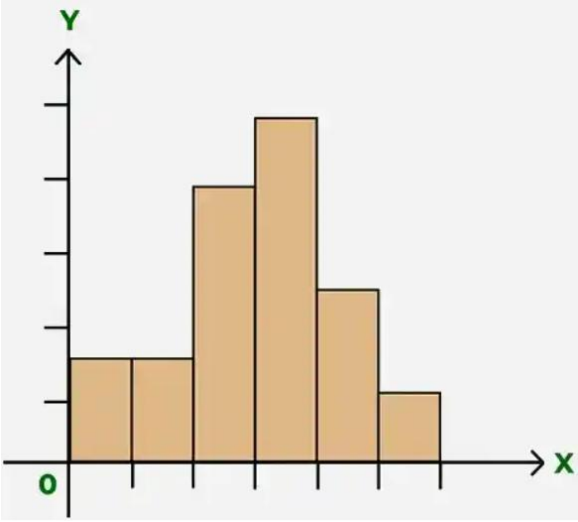
Real Life Example:

1. **Healthcare:** Age distribution of patients.
2. **Education:** Student marks distribution.
3. **Finance:** Income distribution.
4. **Marketing:** Customer purchase frequency.
5. **Machine Learning:** Feature analysis before model training.

Histograms are essential data science tools for visualizing the distribution of continuous numerical data, helping to understand center, spread, and shape. Unlike bar charts, they group continuous data into adjacent intervals (bins) with no spaces, with bar height representing frequency or density.

Key Characteristics

- **Bins:** Continuous, non-overlapping intervals.
- **Frequency:** Height represents how many data points fall in each bin.
- **Density:** Area represents probability density (area under all bars is 1.0).



When we use the Histogram?

A histogram is used in the following situations:

- When the data is numerical.
- To understand the shape and distribution of the data.
- To see whether the data is evenly spread or grouped.
- To compare changes in a process over time.

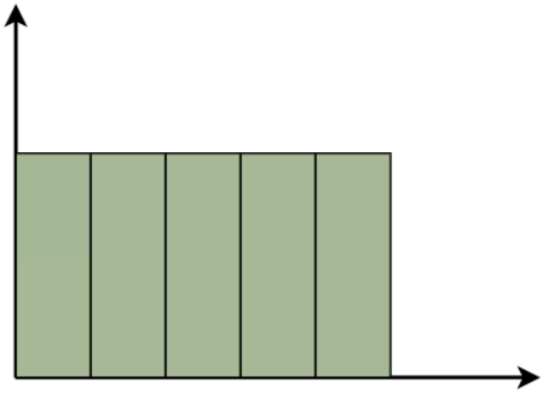
- To compare the results of two or more processes.
- To check whether a process meets customer requirements.
- To present data clearly and quickly for easy understanding.

Types of Histogram

There are various variations of the histograms based on their shapes:

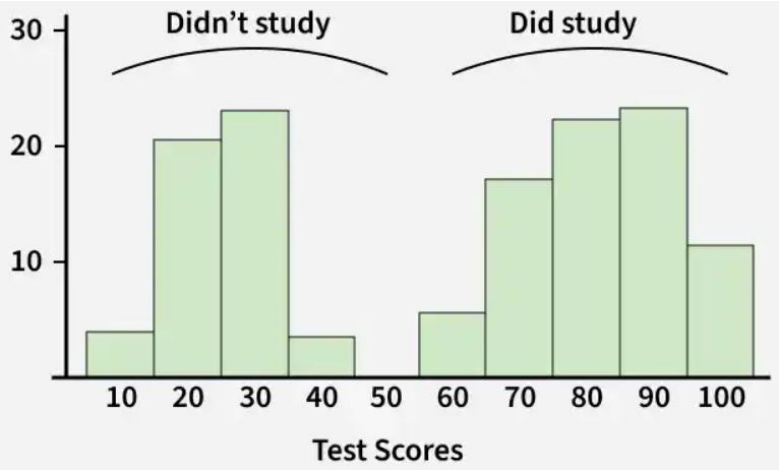
Uniform Histogram

A uniform histogram is a histogram in which all the bars are nearly equal in height, showing that the data is evenly distributed among all class intervals.



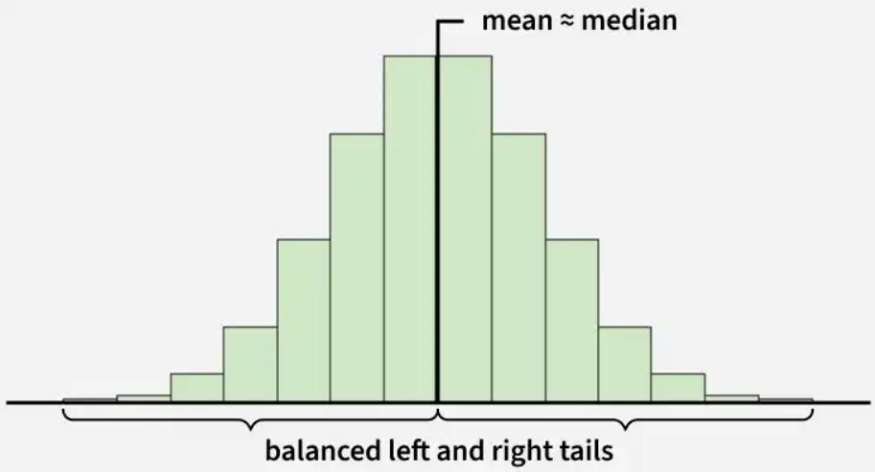
Bimodal Histogram

A histogram is called bimodal if it has two distinct peaks. This shows that the data consists of observations from two different groups or categories, and each group has its own centre or pattern of values.



Symmetric Histogram

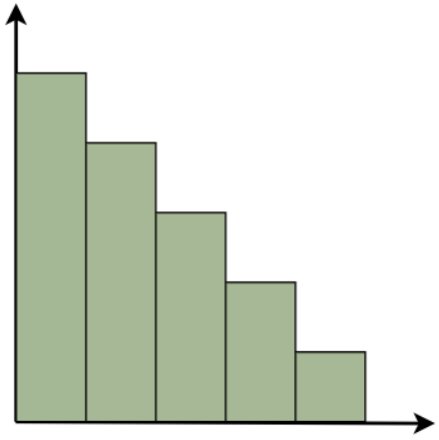
A symmetric histogram is also called a bell-shaped histogram. It is said to be symmetric when a vertical line drawn through the centre divides the histogram into two identical halves. The left and right sides are equal in size and shape, showing a balanced distribution of data.



Right-Skewed Histogram

A right-skewed histogram is a histogram in which the bars extend more towards the right side. This means that most of the data values are on the left side, while a few very large values lie on the right.

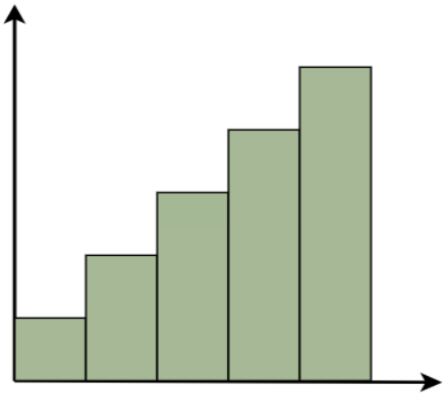
For example: In a histogram showing family incomes, most families may have lower incomes, but a few very rich families stretch the data towards the right.



Left-Skewed Histogram

A left-skewed histogram shows bars that extend towards the left side. This means that most of the data values are on the right side, while a few very low values lie on the left.

For example: In a histogram showing test scores, most students may score high marks, but a few students with very low scores stretch the distribution towards the left.



Advantages of Histograms

1. **Easy to Understand**: Histograms provide a clear visual representation of data distribution.
2. **Shows Frequency Distribution**: They effectively display the frequency of data within different intervals (bins).
3. **Identifies Data Patterns**: Useful for spotting patterns such as skewness, modality, and spread.
4. **Helps Detect Outliers**: Outliers and gaps in data can be easily identified.
5. **Summarizes Large Data Sets**: Condenses large data sets into a simple graphical form.

Disadvantages of Histograms

1. **Loss of Detailed Data**: Exact data values are not visible; only grouped data is shown.
2. **Choice of Bins Affects Interpretation**: Different bin widths or starting points can change the appearance and interpretation.
3. **Not Suitable for Small Data Sets**: Histograms may be misleading or less informative with very small samples.
4. **No Information on Individual Data Points**: Cannot show individual data points or exact values.
5. **Can Be Misleading**: Improper binning or scaling can distort the data representation.

Scatter plot

A Scatter Plot is one of the most important **data visualization techniques** used in **data science, statistics, machine learning, and analytics** to study the **relationship between two numerical variables**.

It represents data points on a **two-dimensional graph**, where each point corresponds to one observation in the dataset.

Scatter plots are mainly used to:

- Identify **relationships (correlation)** between variables

- Detect **patterns and trends**
- Find **outliers**
- Analyze **clusters** in data

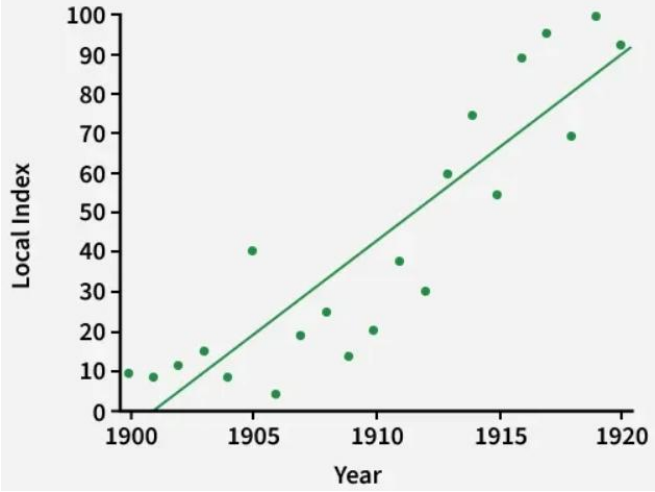
A Scatter Plot is a graphical representation in which **individual data points are plotted on a coordinate plane** to show the relationship between **two continuous numerical variables**.

Each point in the graph represents one observation.

Real-Life example:

1. **Education:** Study hours vs marks
2. **Healthcare:** Exercise time vs blood pressure
3. **Business:** Advertising budget vs sales
4. **Finance:** Risk vs return
5. **Machine Learning:** Feature relationship analysis

A scatter plot, also called a scatter graph or scatter chart, uses dots to describe two different numeric variables. The position of each dot on the horizontal and vertical axes indicates values for an individual data point.



It is generally used to plot the relationship between one independent variable and one dependent variable, where an independent variable is plotted on the x-axis and a dependent variable is plotted on the y-axis so that you can visualize the effect of the independent variable on the dependent variable.

Applications of Scatter Plot

As already mentioned, a scatter plot is a very useful data visualization technique. A few applications of scatter plots are listed below.

- **Correlation Analysis**: A scatter plot is useful in the investigation of the correlation between two different variables. It can be used to find out whether two variables have a positive correlation, negative correlation, or no correlation.
- **Outlier Detection**: Outliers are data points that are different from the rest of the data set. A scatter plot is used to bring out these outliers on the surface.
- **Cluster Identification**: In some cases, scatter plots can help identify clusters or groups within the data.

Scatter Plot Graph

A scatter plot is known by several other names; a few of them are **"scatter chart," "scattergram," "scatter plot," and "XY graph."** A scatter plot is used to visualize a data pair, such that each element gets its axis; generally, the independent one gets the x-axis and the dependent one gets the y-axis.

This kind of distribution makes it easier to visualize the kind of relationship the plotted pair of data is holding. So Scatter Plot is useful in situations when we have to find out the relationship between two sets of data, or in cases when we suspect that there may be some relationship between two variables and this relationship may be the root cause of some problem.

Construction of Scatter Plot

To construct a scatter plot, we have to follow the given steps.

Step 1: Identify the independent and dependent variables

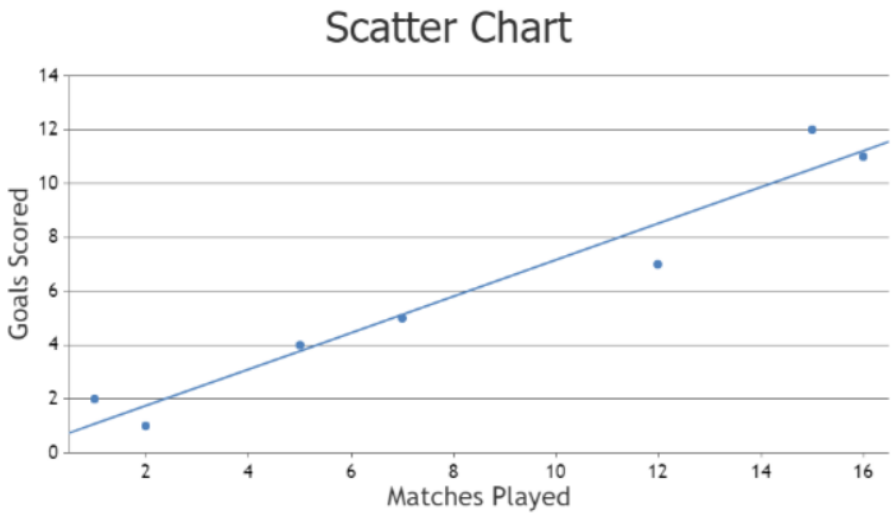
Step 2: Plot the independent variable on the x-axis.

Step 3: Plot the dependent variable on the y-axis.

Step 4: Extract the meaningful relationship between the given variables.

In the following table, a data set of two variables is given.

Matches Played	2	5	7	1	12	15	18
Goals Scored	1	4	5	2	7	12	11



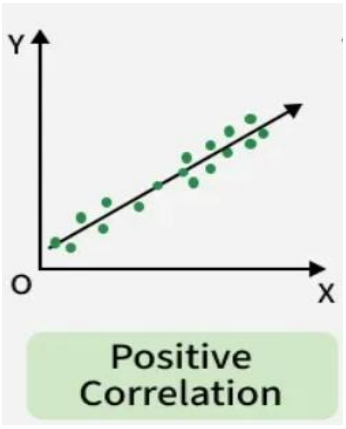
As you can see in the given scatter plot, there is some kind of relationship between the number of matches played and the number of goals scored by a certain player.

Types of Scatter Plot

On the basis of the correlation of two variables, scatter plots can be classified into the following types.

1. Scatter Plot For Positive Correlation

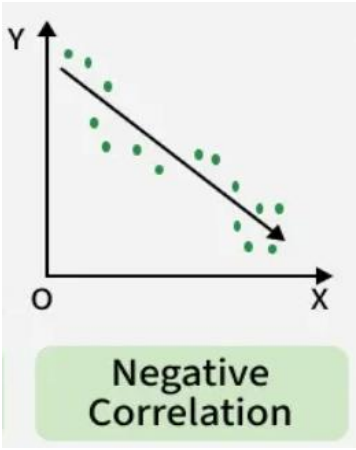
In this type of scatterplot, the value on the y-axis increases on moving from left to right. In more technical terms, if one variable is directly proportional to another, then the scatter plot will show positive correlation. Positive correlation can be further classified into perfect positive, high positive, and low positive.



2. Scatter Plot For Negative Correlation

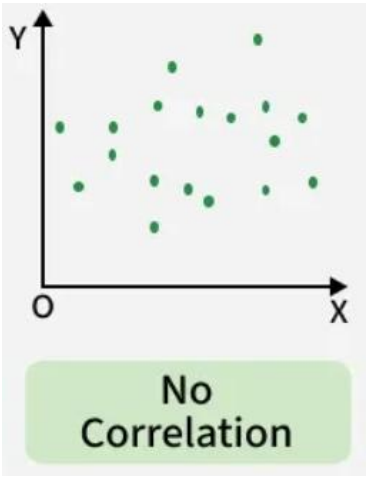
In this type of scatterplot, the value on the y-axis decreases on moving from left to right. In other

words, the value of one variable is decreasing with respect to the other. Positive correlation can be further classified into perfect negative, high negative, and low negative.



3. Scatter Plot For No Correlation

In this type of scatterplot, values are scattered all over the graph. Generally this kind of graph represents that there is no relationship between the two variables plotted on the scatter plot.



Advantages of Scatter Plots

1. **Visualizes Relationships:** Instantly shows correlation type (positive, negative, or no correlation) and strength between variables.
2. **Identifies Outliers:** Makes unusual data points or extreme values immediately obvious.
3. **Detects Patterns/Clusters:** Clearly highlights groups or clusters within the data that might indicate different subgroups.
4. **Supports Linear/Non-linear Analysis:** Visualizes whether the relationship is linear (straight line)

or non-linear (curved), aiding in regression model selection.

5. **Simple Interpretation:** Easy for both technical and non-technical audiences to understand.
6. **Foundation for Modeling:** Often the first step in EDA (Exploratory Data Analysis) before creating complex predictive models.

Disadvantages of Scatter Plots

1. **Limited Dimensions:** Primarily designed for only two variables; adding more makes it hard to read.
2. **Overplotting Large Data:** When thousands of data points exist, points overlap, creating a blob ("overplotting") that hides patterns.
3. **Subjective Interpretation:** Visual analysis can vary; one person might see a trend that another does not.
4. **Correlation vs. Causation:** Shows relationship, but does not prove that one variable causes changes in another.
5. **Only Numeric Variables:** Requires numerical data; cannot be used for purely categorical datasets.
6. **Lack of Precision:** Offers a general visual idea but does not give precise, numerical statistical measurement

Heat map

A Heat Map is a graphical representation of data where values are represented using **different colors**. It is one of the most powerful **data visualization techniques** used in **data science, machine learning, statistics** and **business analytics** to identify **patterns, relationships, trends** and **correlations** in datasets.
In a heat map, data values are displayed as **colors**, where:

1. **Dark/Hot colors** → Higher values
2. **Light/Cool colors** → Lower values

Heat maps help analysts quickly understand complex data by converting numbers into **color-coded visual patterns**.
A Heat Map is a data visualization technique that represents numerical data using **color intensity**, where different colors indicate different magnitudes of values.
It helps in understanding:

1. Correlations

2. Trends
3. Concentrations
4. Relationships among variables

Real-Life Applications

1. **Healthcare:** Disease spread analysis.
2. **E-commerce:** Customer purchase analysis.
3. **Education:** Student performance visualization.
4. **Banking:** Fraud detection.
5. **Website Analytics:** User click tracking.

When to Use Heatmap Visualization

Heatmaps are versatile and can be used in various scenarios:

I. **Website Optimization**: To understand user behavior and optimize webpage design.
II. **Financial Analysis**: To visualize performance metrics and identify areas needing improvement.
III. **Marketing**: To track campaign performance and customer engagement.
IV. **Scientific Research**: To analyze genetic data and other complex datasets.
V. **Geographic Analysis**: Visualizing spatial data such as population density, crime rates, or weather patterns.
VI. **Sports Analytics**: Analyzing player movements, game strategies, or performance metrics.

Types of Heatmaps

Heatmaps can be categorized based on the type of data they visualize and the specific use case. Here are some common types:

1. Website Heatmaps

A website heatmap is a visual tool used to understand how visitors interact with a webpage. It shows which areas receive the most attention using **colors**.

- **Warm colors (Red/Orange)** indicate **high user interaction** or more attention.
- **Cool colors (Blue/Green)** indicate **low interaction** or less attention.

Heatmaps help website owners analyze user behavior such as clicks, scrolling, and mouse movement.

Types of Website Heatmaps

1. **Click Maps** – Show where users click most on a webpage.
2. **Scroll Maps** – Show how far users scroll down a page.

3. **Mouse Tracking Heatmaps** – Track mouse movement to identify attention areas.
4. **Eye-Tracking Heatmaps** – Show where users focus their eyes on the page.



2. Grid Heatmaps

Grid Heatmaps are a data visualization technique used to display data in a **grid or table format**, where each cell is colored according to its value. The color intensity helps in understanding patterns, trends, and relationships in the data.

Key Characteristics

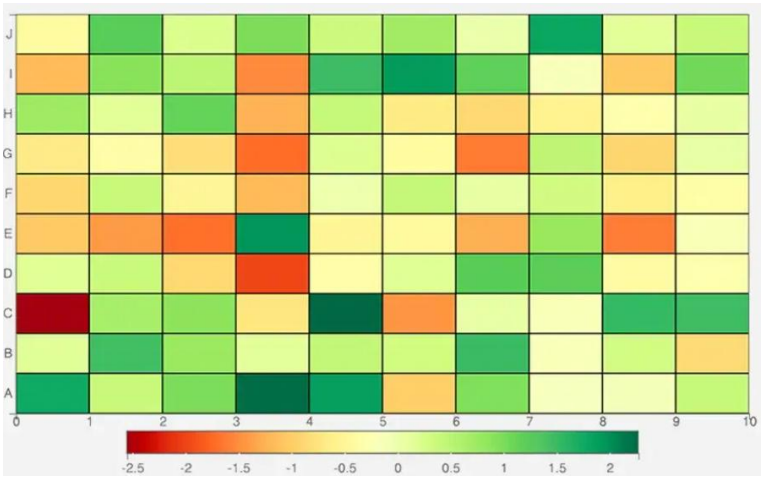
1. **Color Encoding** – Colors represent data values. Darker colors usually indicate **higher values**, while lighter colors indicate **lower values**.
2. **Grid Layout** – Data is arranged in rows and columns, where each cell represents a specific data point.
3. **Data Density** – Can visualize large datasets effectively and identify trends, clusters, or anomalies.
4. **Comparative Analysis** – Helps compare multiple variables at the same time and identify correlations.

Applications

Grid heatmaps are commonly used in:

- **Biology** (gene expression analysis)
- **Finance** (stock market trends)
- **Business analytics** (sales performance comparison)

They are useful for quickly understanding complex data through color patterns.



3. Clustered Heatmaps

Clustered Heatmaps are an advanced type of heatmap that combine **heatmaps with hierarchical clustering** to show relationships between rows and columns. They help group similar data points together, making patterns and correlations easier to identify.

Key Characteristics

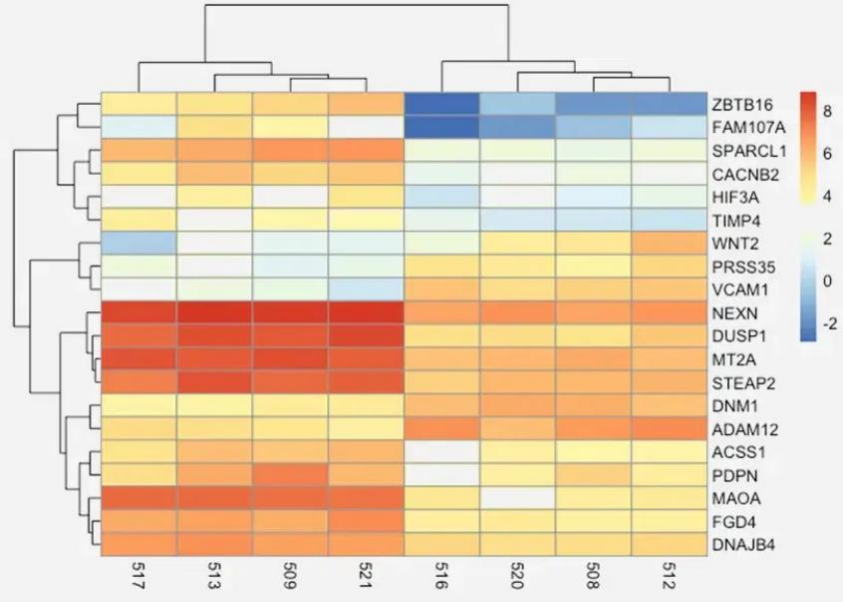
1. **Hierarchical Clustering** – Similar rows and columns are grouped using clustering algorithms and shown with **dendrograms (tree-like diagrams)**.
2. **Color Encoding** – Cell colors represent data values, where darker or stronger colors indicate higher values.
3. **Pattern Identification** – Makes it easier to find trends, similarities, and hidden relationships in data.
4. **Interactive Exploration** – Users can zoom, reorder rows/columns, and explore clusters in detail using software tools.

Applications

Clustered heatmaps are commonly used in:

- **Biology** (gene expression analysis)
- **Data Science** (pattern recognition)
- **Business Analytics** (customer segmentation)

They are useful for visualizing complex datasets and identifying meaningful data relationships quickly.



Advantages of Heat Maps

1. **Rapid Data Interpretation:** Provides an immediate, high-level "bird's-eye view" of large, complex datasets, making it easier to spot trends and patterns at a glance.
2. **Effective for Large Datasets:** Ideal for summarizing massive amounts of information into a compact visual format.
3. **Spotting Patterns and Clusters:** Excel at identifying clusters of similar data points, correlations between variables, and anomalies.
4. **Intuitive Visual Communication:** Color-coded intensities (e.g., darker shades indicating higher concentration) are easily understood by both technical and non-technical audiences.
5. **Versatility in Data Representation:** Effective for various types of analysis, including website user behavior (heatmap tracking), geographical data distribution, and financial performance comparison

Disadvantages of Heat Maps

1. **Loss of Precision/Detail:** The aggregated nature of heat maps can oversimplify data, leading to a loss of granular, precise numerical insights.
2. **Difficulties in Exact Interpretation:** Human visual perception struggles to distinguish between subtle shades of the same hue, making it hard to interpret exact numerical values.
3. **Not Suitable for Dynamic Data:** In web design or app analytics, they often struggle with frequently changing, dynamic layouts, which can produce chaotic "aggregated positional spaghetti".

4. **Subjectivity in Color Interpretation:** The color scale used can lead to subjective or inaccurate interpretations of the data by different viewers.
5. **Misleading Focus:** High-intensity colors can draw attention to insignificant data, masking underlying patterns or leading to premature conclusions

Working with Tableau – Outlier detection

Tableau is a data visualization software that helps users **connect**, **analyze** and **visualize data** from multiple sources to generate meaningful insights.

Tableau is a powerful **Business Intelligence (BI)** and **data visualization tool** used in **data science, analytics** and **business intelligence** to **analyze, visualize** and **interpret data**.

It helps transform **raw data into interactive dashboards, charts, reports** and **visualizations** without requiring extensive programming knowledge.

Tableau supports:

1. Data analysis
2. Dashboard creation
3. Reporting
4. Data exploration
5. Predictive analytics

Features of Tableau

1. Drag-and-Drop Interface: Users can create visualizations without coding.

2. Interactive Dashboards: Allows dynamic filtering and interaction.

3. Real-Time Data Analysis: Supports live data connections.

4. Multiple Data Source Integration: Can connect with: Excel, CSV files, SQL databases, Google Sheets, Cloud databases

5. Big Data Support: Can process very large datasets.

Applications of Tableau

1. Healthcare
2. Banking
3. Education

4. Marketing
5. Business analytics
6. Data science

Values in Tableau

There are two types of values in the tableau:

1. **Dimensions:** Values that are discrete(which can not change with respect to time) in nature called Dimension in tableau. Example: city name, product name, country name.
2. **Measures:** Values that are continuous(which can change with respect to time) in nature called Measure in tableau. Example: profit, sales, discount, population.

Advantages of Tableau

1. **Quick calculation-** All the calculations on the tableau done by the backend, so it is relatively faster than any other tool.
2. **Interactive dashboards**- Tableau dashboards are very interactive and easy to draw.
3. **No manual calculation-** All the calculations are done by the tableau only. There is no manual calculation, but in some specific cases, we used calculated fields for calculation.
4. **A large amount of data-** Tableau can handle a large amount of data. Different types of visualization can be created with a large amount of data without impacting the performance of the dashboards.

Disadvantages of Tableau

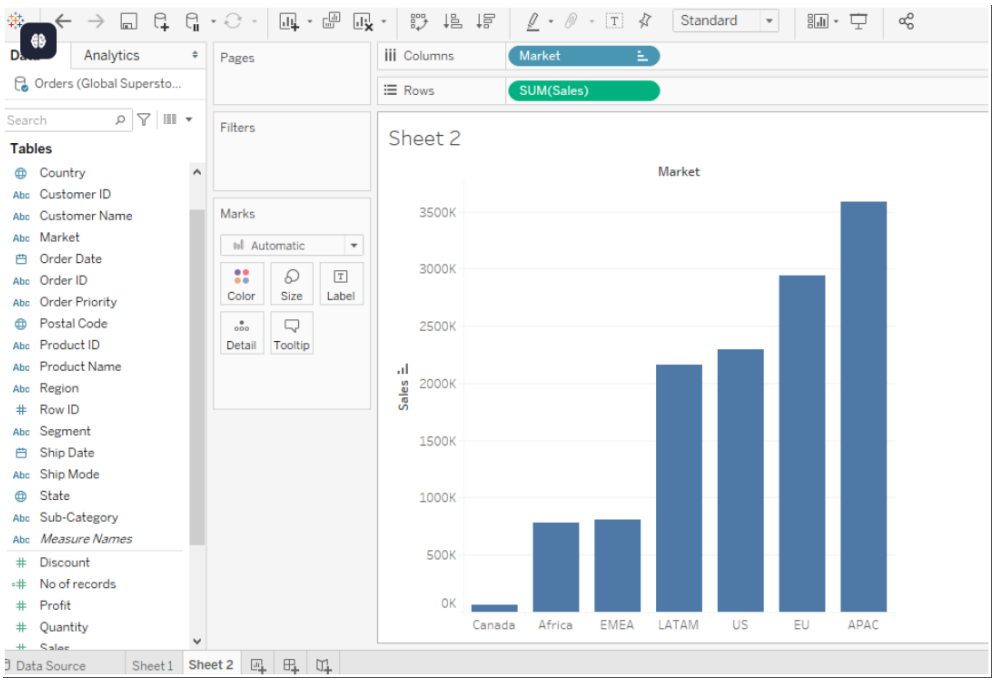
1. **High Cost-** tableau is a paid tool for visualization, and it is a reason why people are not using tableau so much.
2. **Static and single value parameters-** Tableau's parameters are static and always single value can be selected using a parameter. Whenever the data gets changed, these parameters need to be updated manually every time.
3. **Limited Data Preprocessing-** Tableau is strictly a visualization tool. Tableau Desktop allows you to do very basic preprocessing.

Visualizations in tableau

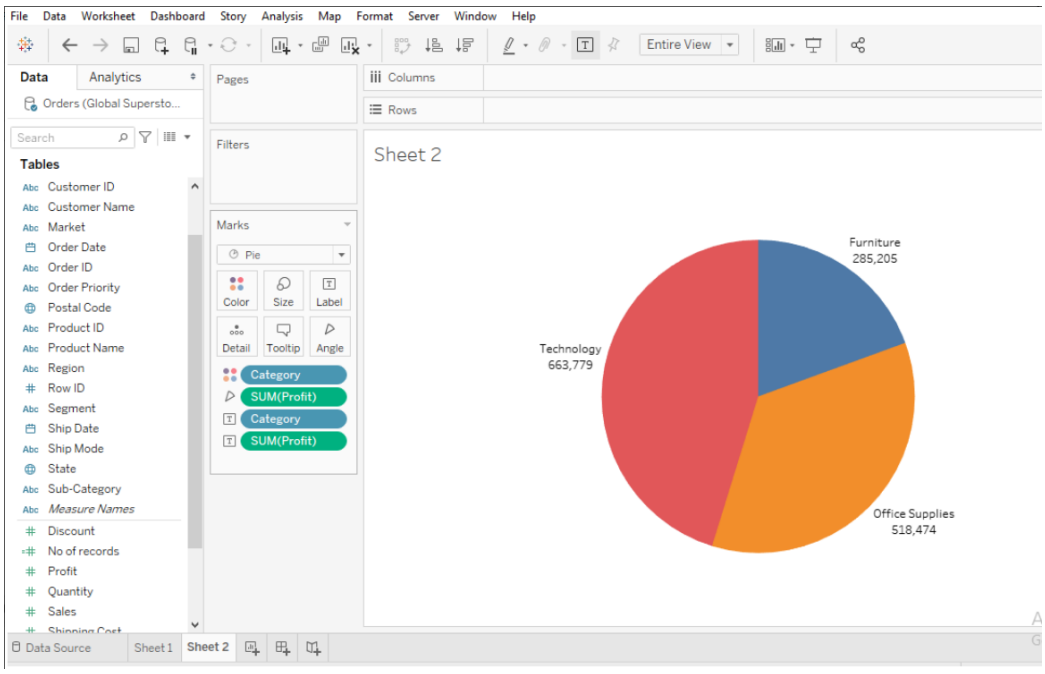
There are a lot many charts available in the tableau. Some of them are:

- **Bar chart:** For the bar chart, we need 0 or more dimensions and 1 or more measures. To create a

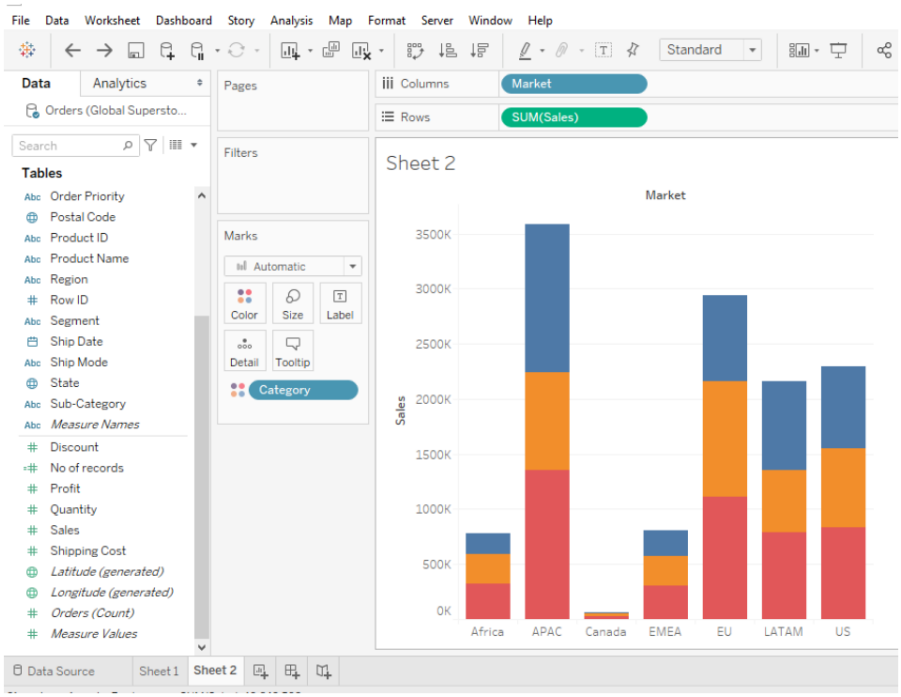
bar chart, simply select the desired dimensions and measures and then select the bar chart on the **SHOW ME** section, or drag the dimensions in a column and measure in a row.



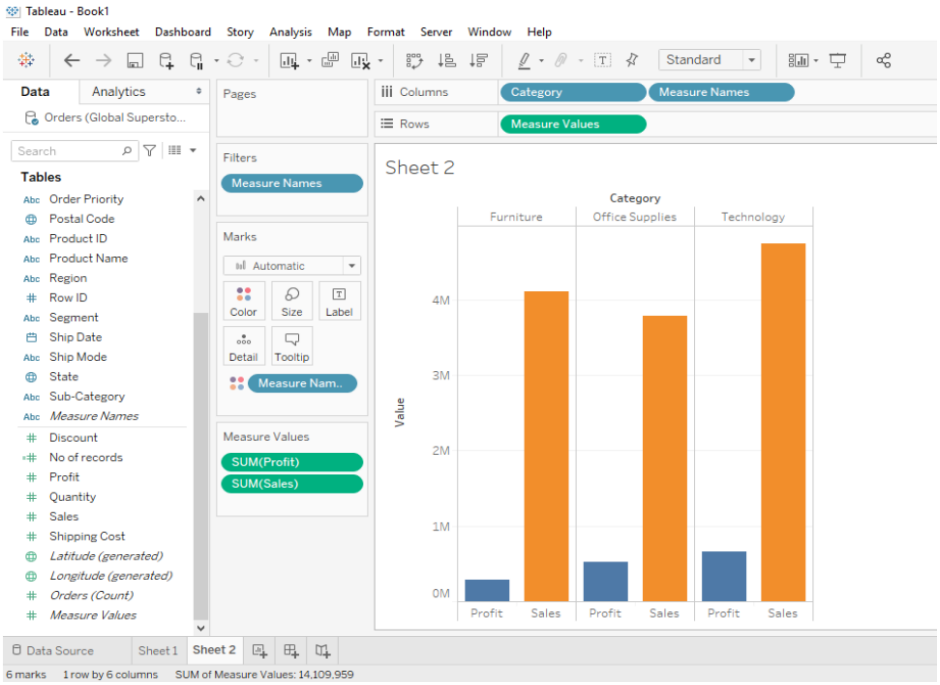
- **Pie chart:** For the pie chart, we need 1 or more dimensions and 1 or 2 measures. To create a pie chart, simply select the desired dimensions and measures and then select Pie chart on the **SHOW ME** section.



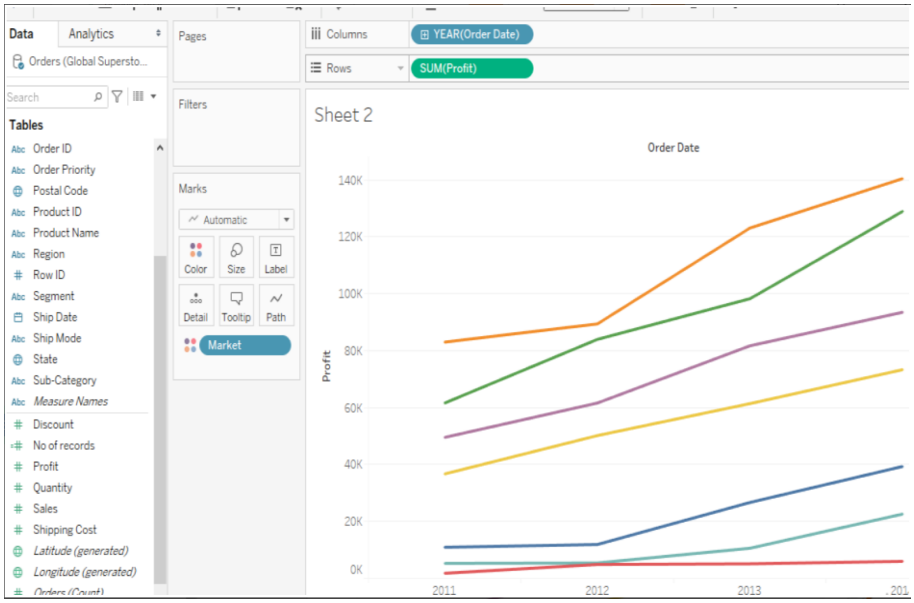
- **Stacked bar chart:** For the stacked bar chart, we need 1 or more dimensions and 1 or more measures. To create a stacked bar chart, simply select the desired dimensions and measures and then select the stacked bar chart on the **SHOW ME** section.



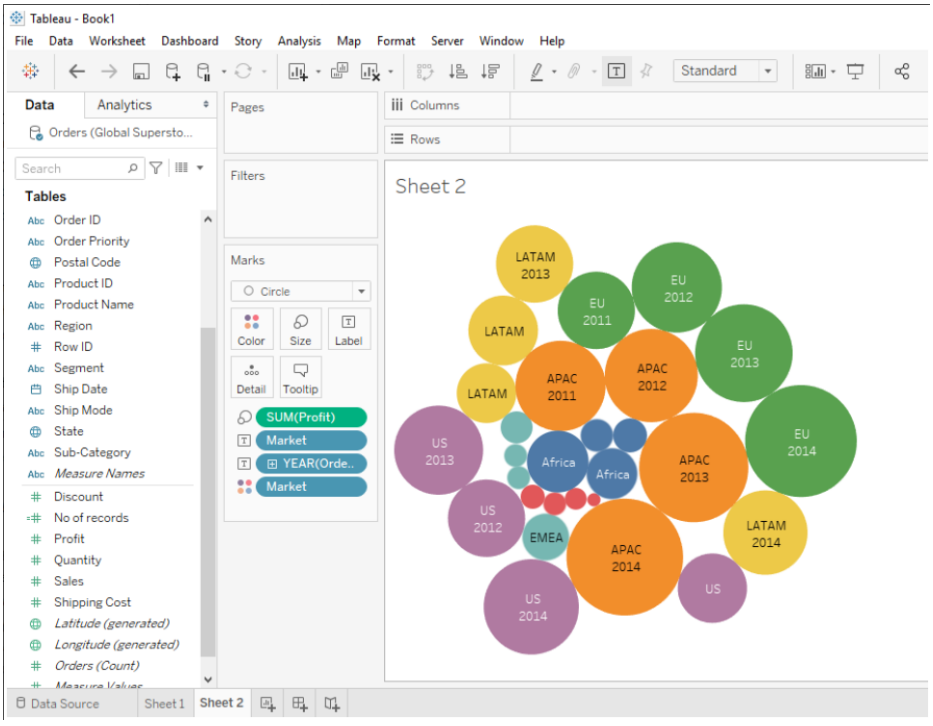
- **Side-by-side bar chart:** For the side-by-side bar chart, we need 1 or more dimensions and 1 or more measures. To create a side-by-side bar chart simply select the desired dimensions and measures and then select a side-by-side bar chart on the **SHOW ME** section.



- **Line chart:** For the line chart, we need 1 date, 0 or more dimensions, and 1 or more measures. To create a line chart, simply select the desired dimensions and measures and then select the line chart on the **SHOW ME** section.



- **Packed bubbles chart:** For the packed bubbles chart, we need 1 or more dimensions and 1 or 2 measures. To create a packed bubble chart simply select the desired dimensions and measures and then select the packed bubbles chart on the **SHOW ME** section.



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Outlier Detection

In data science, Outlier Detection is the process of identifying unusual, abnormal or extreme data points that significantly differ from the rest of the dataset.

Outliers may occur because of:

- Data entry errors
- Experimental errors
- Fraudulent activities
- Rare events

An Outlier is a data point that lies significantly far away from other observations in a dataset.

Example:

Dataset: 10, 12, 15, 18, 20, 22, 25, 150.

Here, 150 is an outlier.

Applications

1. Fraud detection
2. Medical diagnosis
3. Cybersecurity
4. Finance
5. Machine learning

Types of Outliers

1. Global Outlier: Very different from entire dataset.

Example:

Salary = ₹10 lakh among ₹20,000 salaries.

2. Contextual Outlier: Abnormal in specific situation.

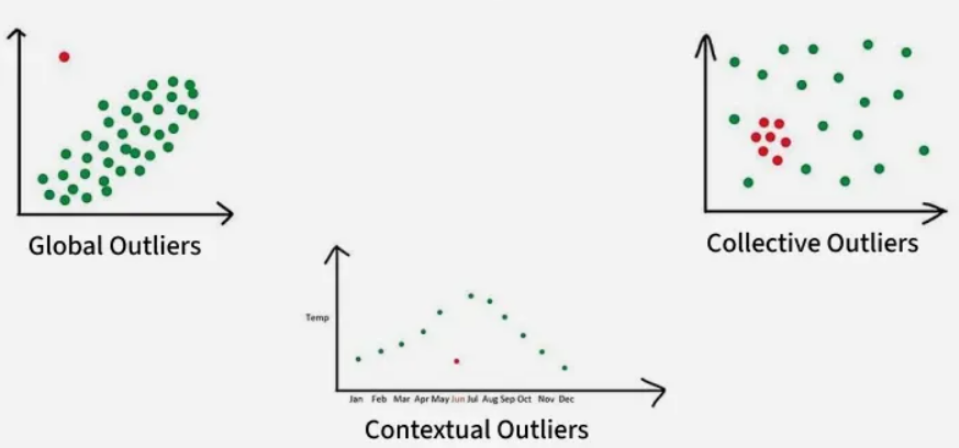
Example:

30°C temperature in winter.

3. Collective Outlier: Group of unusual observations.

Example:

Suspicious bank transactions.



Methods for Outlier Detection

Outlier detection is a critical task in data analysis, crucial for ensuring the quality and reliability of conclusions drawn from data. Different techniques are tailored for varying data types and scenarios, ranging from statistical methods for general data sets to specialized algorithms for spatial and temporal data. Some techniques are:

Standard Deviation Method

Standard Deviation Method is based on the assumption that the data follows a normal distribution. Data points outside of three standard deviations from the mean are considered outliers.

It is commonly used for univariate data analysis where the distribution can be assumed to be approximately normal.

- **Step 1**: Calculate the average and standard deviation of the data set, if applicable.
- **Step 2**: Define the lower and upper bounds for outliers.
- **Step 3**: Identify outliers as data points that fall outside these bounds:
-

Example: Dataset: [1, 2, 2, 3, 1, 3, 10]. Find an outlier using the Standard Deviation Method.

$$\text{Mean, } \mu = \frac{(1+2+2+3+1+3+10)}{7} = \frac{22}{7} \approx 3.14$$

$$\text{Standard Deviation, } s \approx \sqrt{\frac{(1-3.14)^2+(2-3.14)^2+(2-3.14)^2+(3-3.14)^2+(1-3.14)^2+(3-3.14)^2+(10-3.14)^2}{7}} \approx 2.90$$

Lower and upper bounds for outliers:

- *Lower bound = 3.142857 - 2 X 2.917 3.142857 - 5.834 = -2.691*
- *Upper bound = 3.142857 + 2 X 2.917 3.142857 + 5. 834 = 8.977*

Any values outside [-2.691, 8.977] is an outlier, thus 10 is an outlier.

IQR Method

The Interquartile Range (IQR) method focuses on the spread of the middle 50% of data. It calculates the IQR as the difference between the 75th and 25th percentiles of the data and identifies outliers as those points that fall below 1.5 times the IQR below the 25th percentile or above 1.5 times the IQR above the 75th percentile. This method is robust to outliers and does not assume a normal distribution.

- Step 1: Find Q1(25th percentage) and Q3(75th percentage)
- Step 2: IQR = Q3 - Q1.
- Step 3: Find Lower Bound: Q1 - 1.5 × IQR and Upper Bound Lower Bound: Q1 - 1.5 × IQR

It is suitable for datasets with skewed or non-normal distributions. Useful for identifying outliers in datasets where the spread of the middle 50% of the data is more relevant than the mean and standard deviation.

Example: Dataset *X* = {3,5,7,9,11,13,30}, find outlier using the IQR method.

- *Q1 (25th percentile): Median of first half = Median of [3, 5, 7] = 5*
- *Q3 (75th percentile): Median of second half = Median of [11, 13, 30] = **13***

IQR = Q3 – Q1 = 13 – 5 = 8
Lower Bound: Q1 - 1.5 × IQR = 5 - 1.5 × 8 = 5 - 12 = -7
Upper Bound: Q3 + 1.5 × IQR = 13 + 1.5 × 8 =13 + 12 = 25
Therfore the interval is -7 to 25. 30 lies outside the interval, therefore is an outlier.

Z-Score Method

The Z-score method calculates the number of standard deviations each data point is from the mean. A Z-score threshold is set, commonly 3, and any data point with a Z-score exceeding this threshold is considered an outlier. This method assumes a normal distribution and is sensitive to extreme values in small datasets.

- Step 1: Calculate the mean.
- Step 2: Compute Standard Deviation
- Step 3:Calculate z-scores
- Step 4: Apply Threshold Rule: Mild outlier: |Z| > 2 and Extreme outlier: |Z| > 3

Suitable for datasets with large sample sizes and where the underlying distribution of the data can be reasonably approximated by a normal distribution.

Example: *X* = {4,5,5,6,7,8,20}, find outlier using the Z-score method.

Mean, $\bar{X} = \frac{4+5+5+6+7+8+20}{7} = \frac{55}{7} \approx 7.86$

Standard Deviation, $s \approx \sqrt{\frac{(4-7.86)^2+(5-7.86)^2\times2+(6-7.86)^2+(7-7.86)^2+(8-7.86)^2+(20-7.86)^2}{6}} \approx 5.36$

Z-scores: $Z_i = \frac{X_i-7.86}{5.36}$

*Z-score for all data points: 4: -0.72, 5: -0.53, 5: -0.53, 6: -0.35, 7: -0.16, 8: 0.03, 20: **2.26***

Threshold Z > 2 := 20 is an outlier since |2.26| > 2

Outlier Detection

Outlier detection is a process of identifying observations or data points that significantly deviate from the majority of the data.

Outliers can distort statistical analyses, leading to erroneous conclusions and misleading interpretations. When calculating means, medians, or standard deviations, outliers can exert disproportionate influence, skewing the results and undermining the validity of the analysis. By

detecting and appropriately addressing outliers, analysts can mitigate the impact of these anomalies on statistical measures, ensuring that the insights drawn from the data are representative and accurate.

Detecting outliers is critical for numerous reasons:

1. **Improving Accuracy**: Removing or accurately handling outliers enhances the performance and predictability of data models.
2. **Fraud Detection**: Outliers can be symptomatic of fraudulent activity, especially in financial or transaction data.
3. **Data Quality**: Regular outlier detection is crucial to maintain the integrity and quality of data, which in turn affects the decision-making processes based on this data.
4. **Model Performance**: Outliers can significantly impact the performance of statistical models, machine learning algorithms, and other analytical techniques. By identifying and handling outliers appropriately, we can improve the robustness and accuracy of these models.
5. **Insight Generation**: Outliers may represent unique or interesting phenomena in the data. Identifying and analyzing outliers can lead to valuable insights, such as detecting emerging trends, understanding rare events, or uncovering potential opportunities or threats.

Challenges with Outlier Detection

Detecting outliers effectively poses several challenges:

- **Determining the Threshold**: Deciding the correct threshold that accurately separates outliers from normal data is critical and difficult.
- **Distinguishing Noise from Outliers**: In datasets with high variability or noise, it can be particularly challenging to differentiate between noise and actual outliers.
- **Balancing Sensitivity**: An overly aggressive approach to detecting outliers might eliminate valid data, reducing the richness of the dataset.

Data Balancing

In data science and machine learning, Data Balancing is the process of adjusting an imbalanced dataset so that all classes are represented equally or fairly.

An imbalanced dataset occurs when one class contains significantly more observations than another class.

Data Balancing is the technique of modifying a dataset to ensure that all target classes have a balanced or nearly balanced number of observations.

Its goal is to **reduce bias and improve model performance**.

For example:

Class	Count
Yes	950
No	50

Applications of Data Balancing

1. **Fraud Detection**: Credit card fraud.
2. **Medical Diagnosis**: Rare disease prediction.
3. **Spam Detection**: Spam vs non-spam emails.
4. **Cyber security**: Intrusion detection.
5. **Customer Churn Prediction**: Identifying leaving customers.

Importance of Data Balancing

Data balancing helps in:

1. Improving prediction accuracy
2. Preventing biased models
3. Better minority class prediction
4. Enhancing model reliability
5. Improving classification performance

Imbalanced Dataset Problem

An imbalanced dataset can cause:

1. Biased Model: Model favors majority class.

2. Poor Accuracy for Minority Class: Minority predictions become weak.

3. Overfitting: Model memorizes dominant class.

4. Misleading Accuracy

Example: If 95% data is class A:

Model predicting only A gives 95% accuracy but performs poorly.

Types of Imbalanced Data

1. **Binary Class Imbalance:** Only two classes exist.

   Example: Spam / Not Spam

2. **Multi-Class Imbalance:** Multiple classes are uneven.

   Example:

Grade	Count
A	100
B	500
C	50

Data Balancing Workflow

Raw Dataset

↓

Check Class Distribution

↓

Imbalanced?

↓

Choose Method

(Over/Under/Hybrid)

↓

Balanced Dataset

↓

Train ML Model

Evaluation Metrics for Imbalanced Data

Accuracy alone is misleading.

Important metrics:

1. Precision

Precision = TP / (TP + FP)

2. Recall

Recall = TP / (TP+FN)

3. F1 Score

F1Score = 2 × (Precision × Recall) / (Precision + Recall)

4. ROC-AUC

Measures classification quality.

Methods of Data Balancing

There are mainly three approaches:

A. Oversampling

Definition: Oversampling increases the number of observations in the **minority class**.

It duplicates or creates synthetic samples.

Example: Before Balancing

Class	**Count**
Yes	900
No	**100**

After Oversampling

Class	**Count**
Yes	900
No	**900**

Types of Oversampling:

1. Random Oversampling: Duplicates minority data randomly.

Example: Copying minority records.

2. SMOTE (Synthetic Minority Over-sampling Technique): Generates **synthetic/artificial data points** instead of duplication.

Working of SMOTE

1. Select minority sample
2. Find nearest neighbors
3. Generate synthetic values

B. Under sampling

Definition: Under sampling reduces the number of observations in the majority class.

Example: Before Balancing

Yes	**900**
No	100

After Oversampling

Yes	**100**
No	100

Types of Under sampling:

1. Random Under sampling: Randomly removes majority data.

2. Cluster Centroid: Uses clustering to reduce majority samples.

C. Hybrid Sampling

Definition: Combination of Oversampling and Under sampling

Example: SMOTE + Under sampling

Advantage

- Balances performance and efficiency.

References:

Textbook

1. Ethem Alpaydin, Introduction to Machine Learning, Fourth Edition, MIT Press, 2020
2. R for Data Science by Hadley Wickham & Garrett Grolemund (Comprehensive R programming guide)
3. Python for Data Analysis by Wes McKinney (Covers Pandas, NumPy, and Jupyter, written by the creator of Pandas)
4. Pattern Recognition and Machine Learning by Christopher M. Bishop (Covers deep math and algorithms)

Reference book

1. Hadley Wickham, Garrett Grolemund, R for data science : Import, Tidy, Transform, Visualize, And Model Data Paperback, 2017.
2. Han, J., Kamber, M., Pei, J. Data mining concepts and techniques. Morgan Kaufmann. 2011
3. James, G., Witten, D., T., Tibshirani, R. An Introduction to statistical learning with applications in R. Springer. 2013

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